

City-conditioned urban scene atlas

Regular fabric

Informal small-roof fabric

Mixed complex fabric

Douala



Yaounde



Garoua



Dakar



Kigali



Dar es Salaam

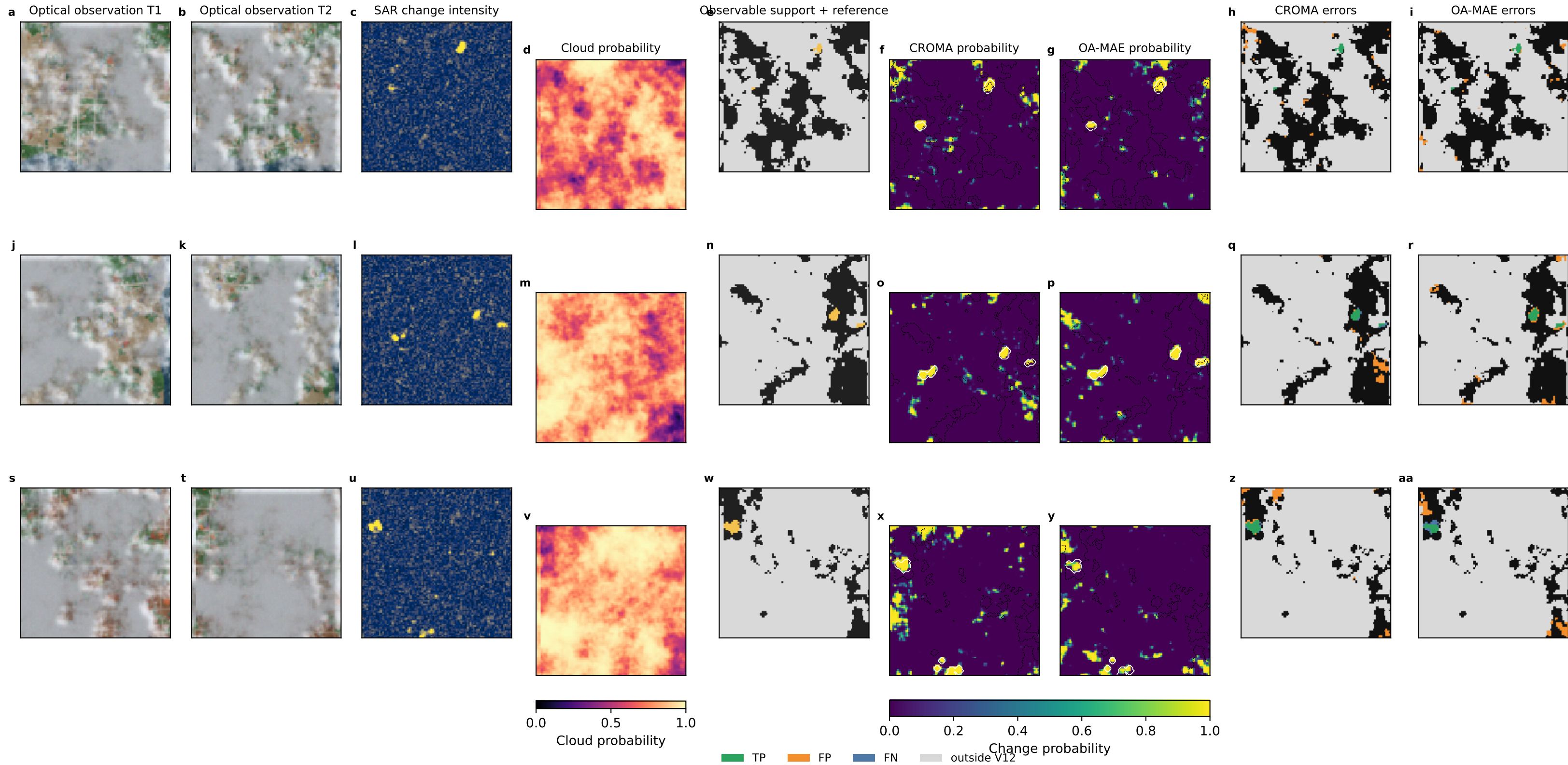


OA-MAE qualitative comparison under severe cloud burden

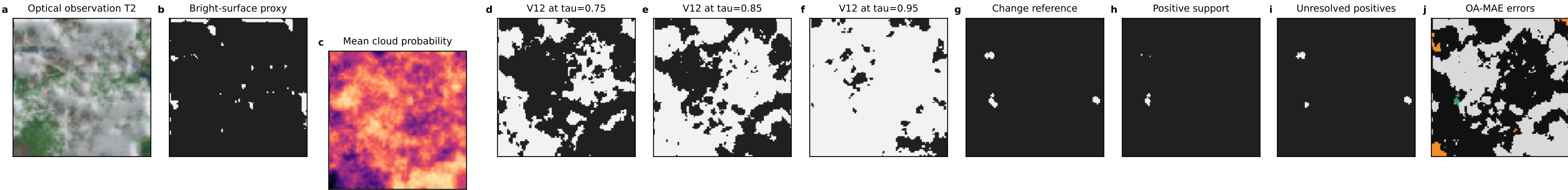
Median positive gain
Dar es Salaam | SCN_dar_0066
cloud 0.741 | coverage 0.343
IoU CROMA 0.236, OA-MAE 0.263 | dIoU +0.027

Upper-quartile gain
Dar es Salaam | SCN_dar_0065
cloud 0.790 | coverage 0.238
IoU CROMA 0.270, OA-MAE 0.316 | dIoU +0.046

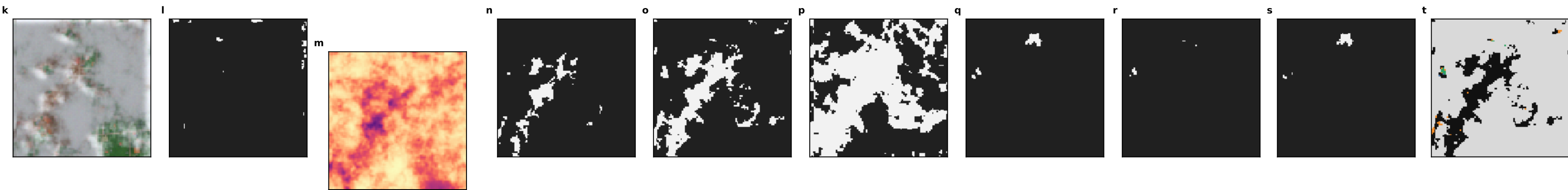
Largest regression
Kigali | SCN_kig_0066
cloud 0.843 | coverage 0.132
IoU CROMA 0.333, OA-MAE 0.187 | dIoU -0.147



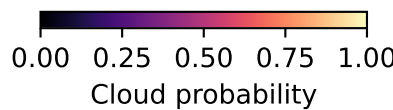
Observability, threshold sensitivity, and unresolved-positive diagnostics



Bright-surface sensitivity
Douala | SCN_dou_0064
coverage .75/.85/.95: 0.37/0.57/0.89
positive support: 0.22/0.29/0.86

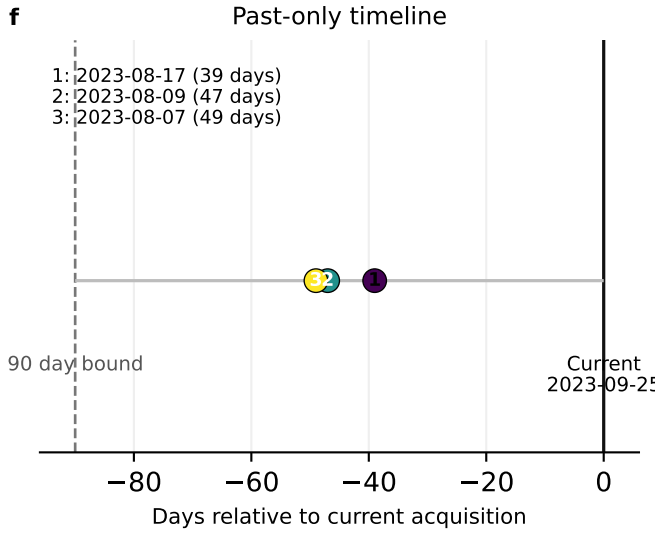
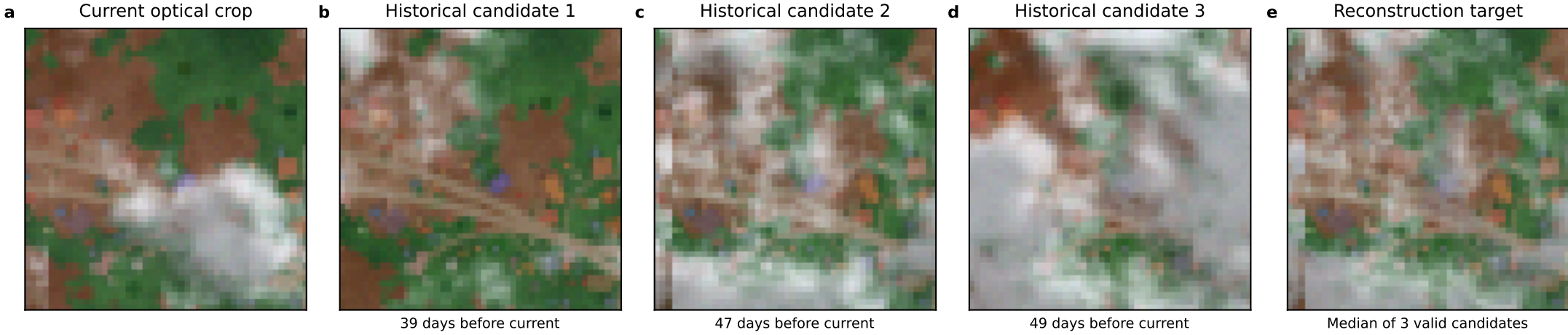


Unresolved-positive burden
Kigali | SCN_kig_0069
coverage .75/.85/.95: 0.06/0.16/0.49
positive support: 0.02/0.17/0.77

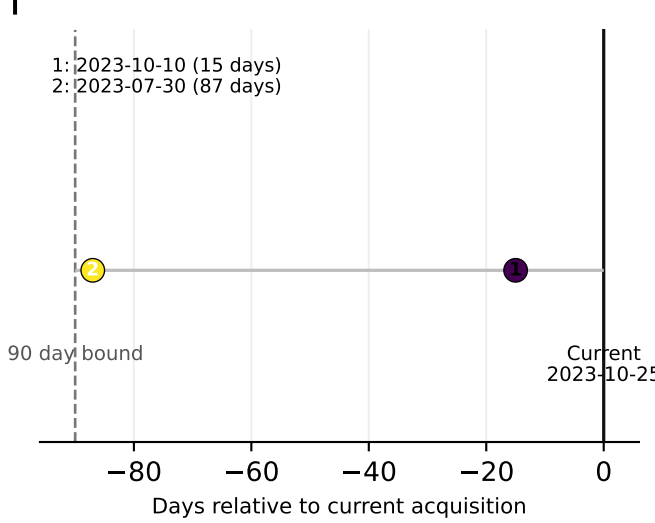
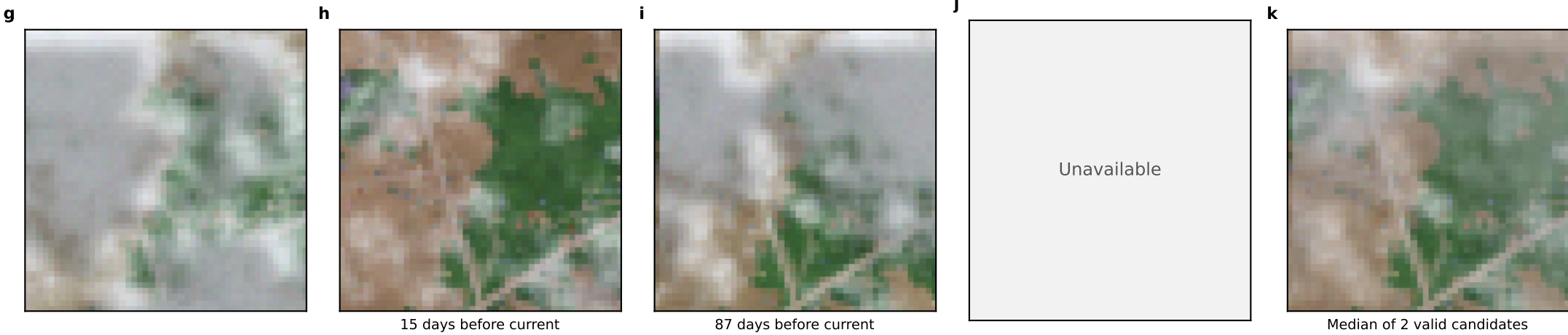


Past-only target retrieval and fallback diagnostics

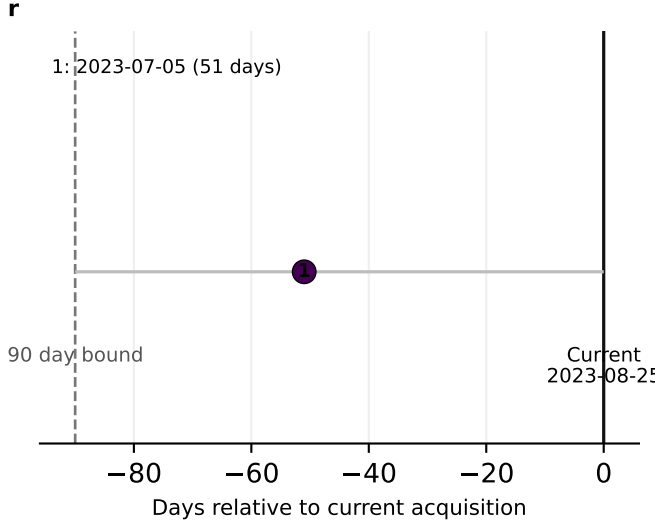
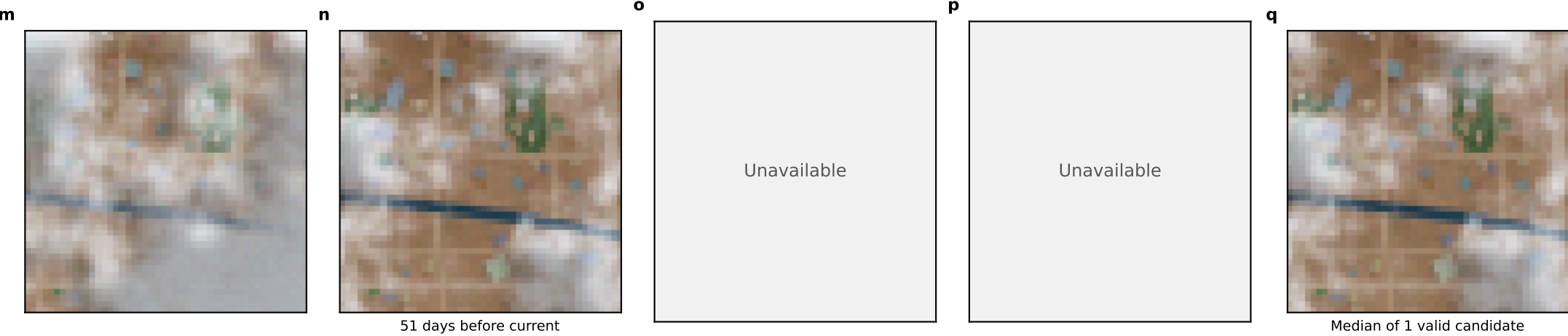
Three valid historical targets
Yaounde | SCN_yao_0065 | tok_000
Season wet | safety weight 0.718



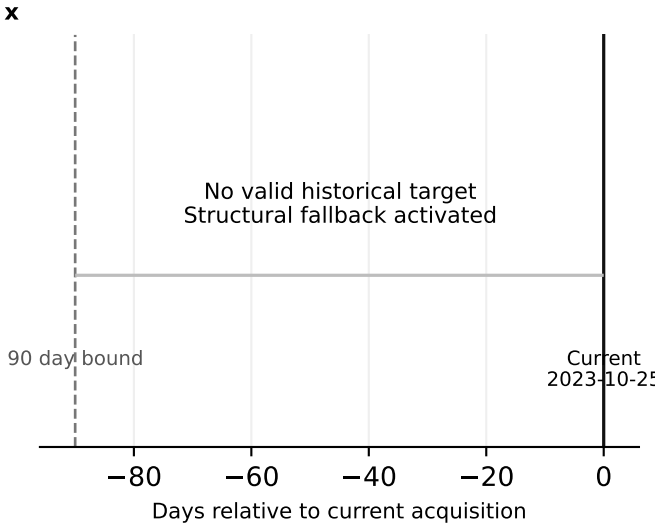
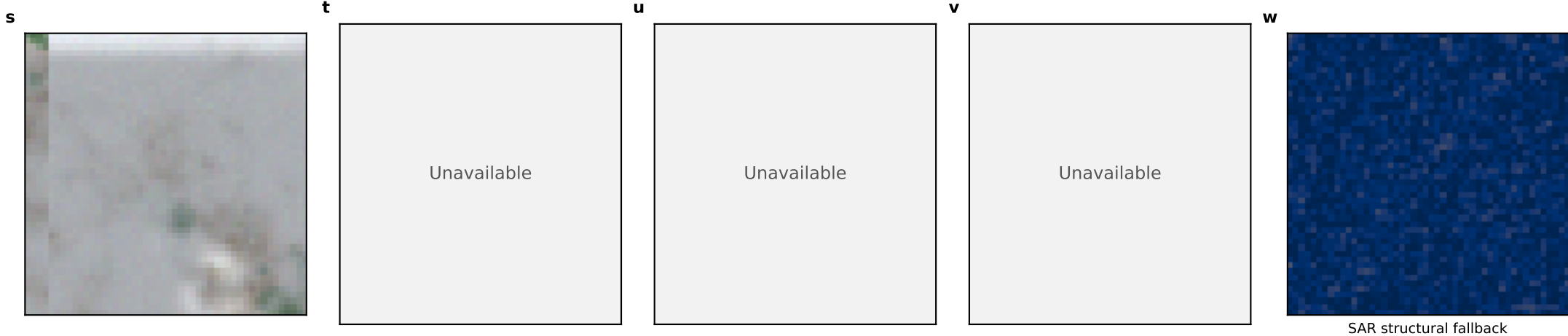
Two valid historical targets
Dar es Salaam | SCN_dar_0058 | tok_000
Season wet | safety weight 0.694



One valid historical target
Garoua | SCN_gar_0069 | tok_000
Season dry | safety weight 0.694



Structural fallback
Dakar | SCN_dak_0058 | tok_000
Season wet | safety weight 0.460

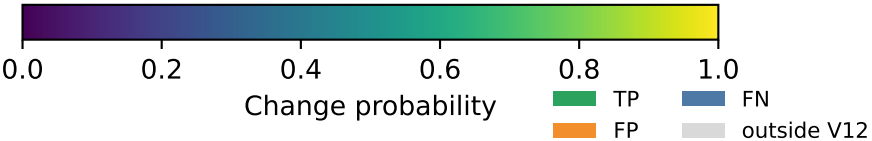
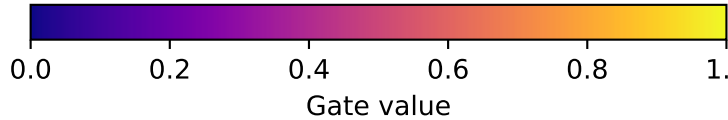
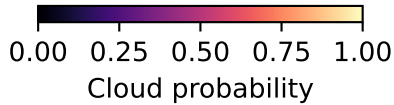
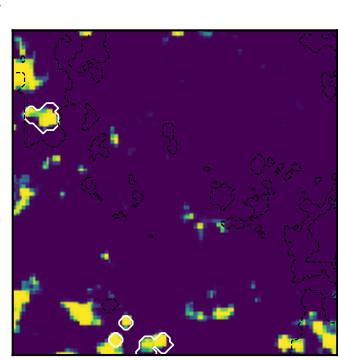
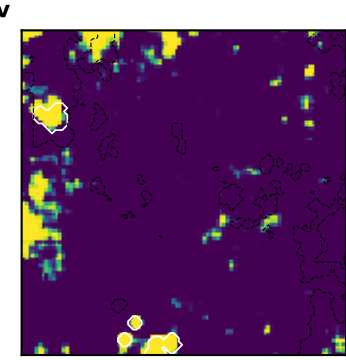
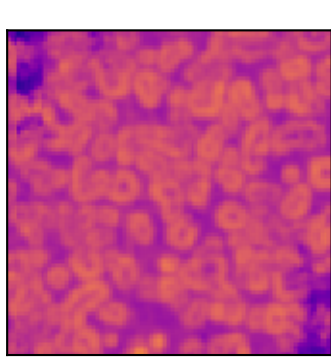
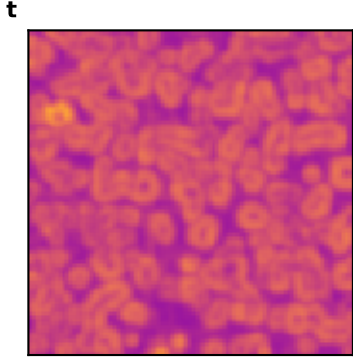
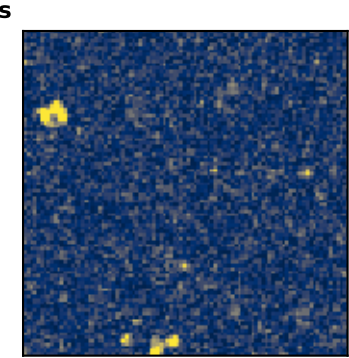
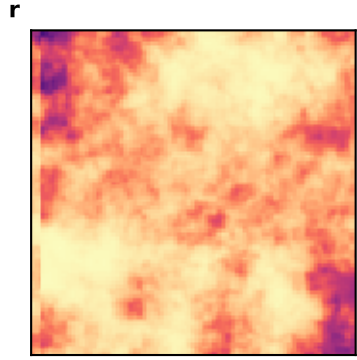
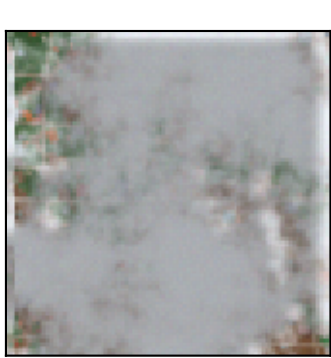
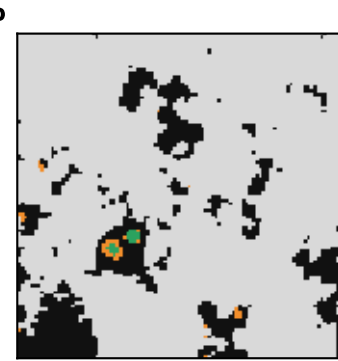
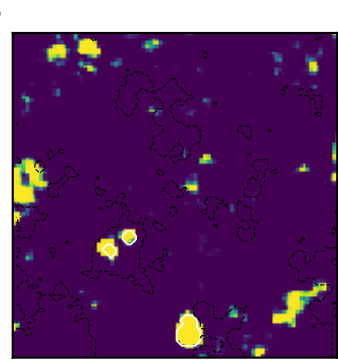
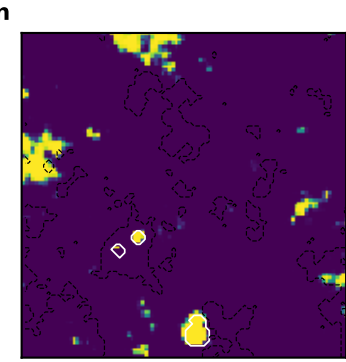
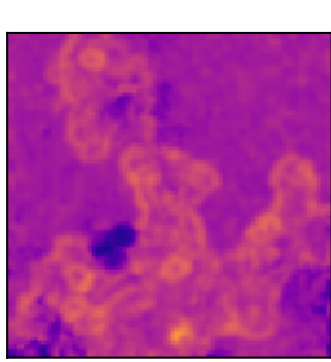
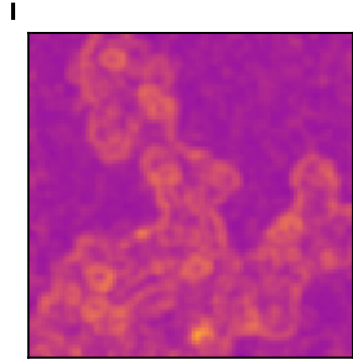
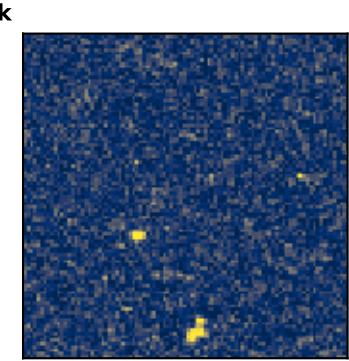
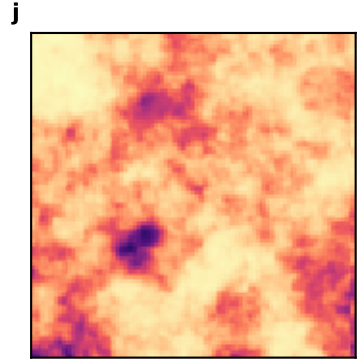
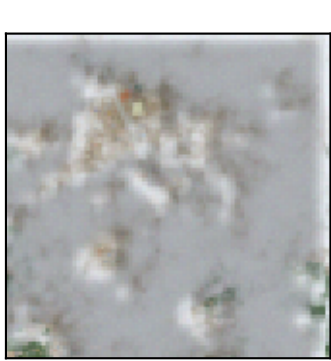
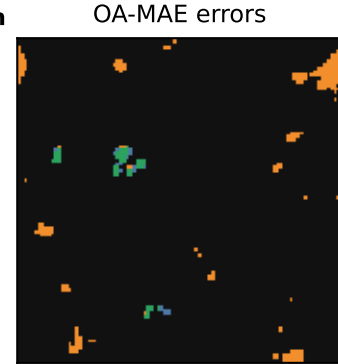
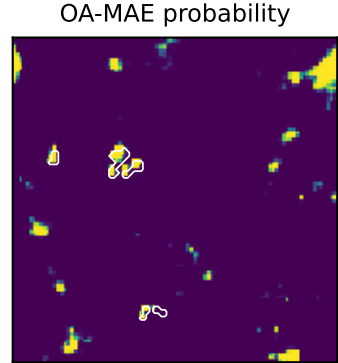
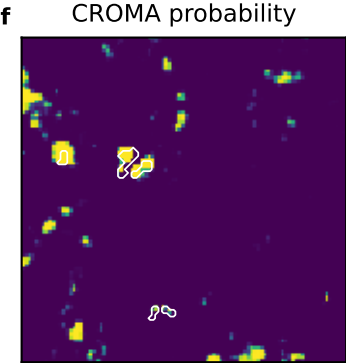
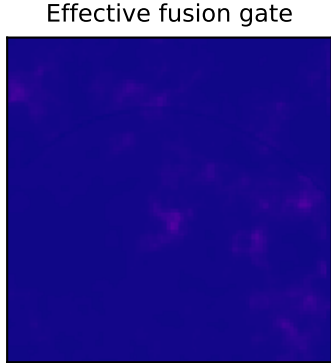
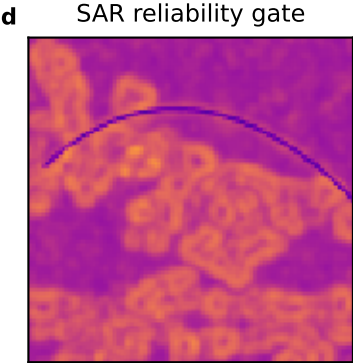
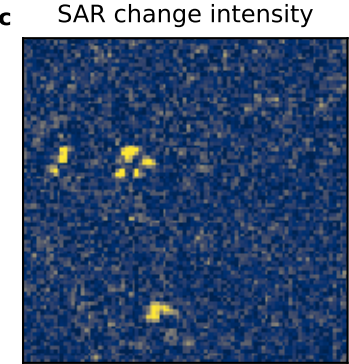
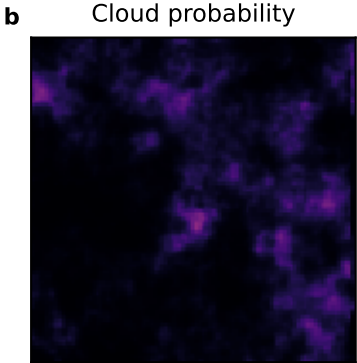
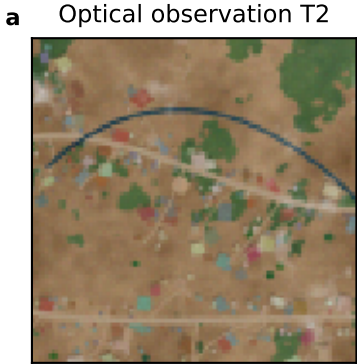


Mechanism diagnostics for observability-conditioned SAR-optical fusion

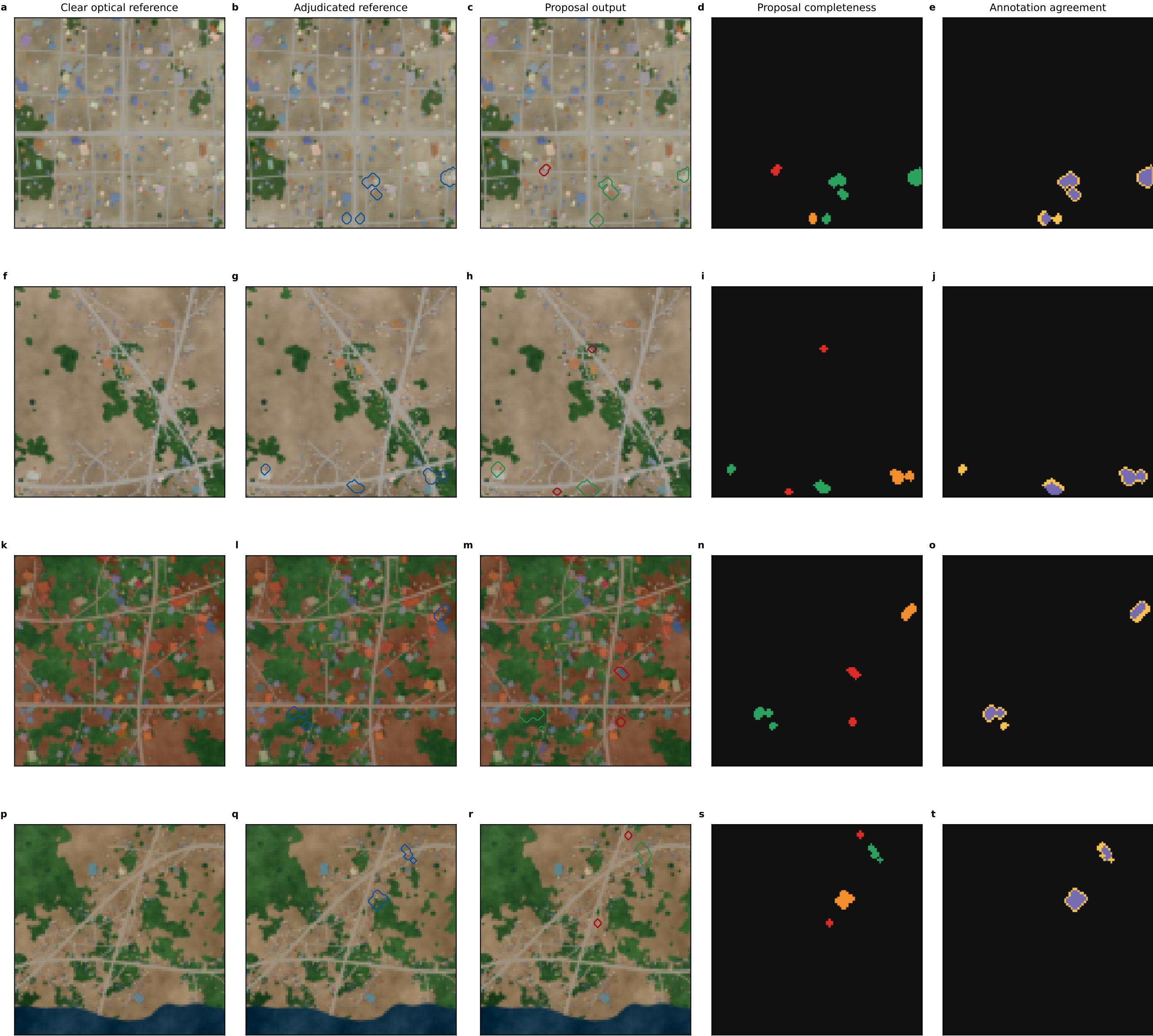
Clear optical evidence
Garoua | SCN_gar_0059
cloud 0.060 | SAR gate 0.466
effective gate 0.013 | dIoU -0.004

Severe cloud - largest OA-MAE gain
Dakar | SCN_dak_0067
cloud 0.822 | SAR gate 0.422
effective gate 0.391 | dIoU +0.180

Severe cloud - largest OA-MAE regression
Kigali | SCN_kig_0066
cloud 0.843 | SAR gate 0.507
effective gate 0.475 | dIoU -0.147

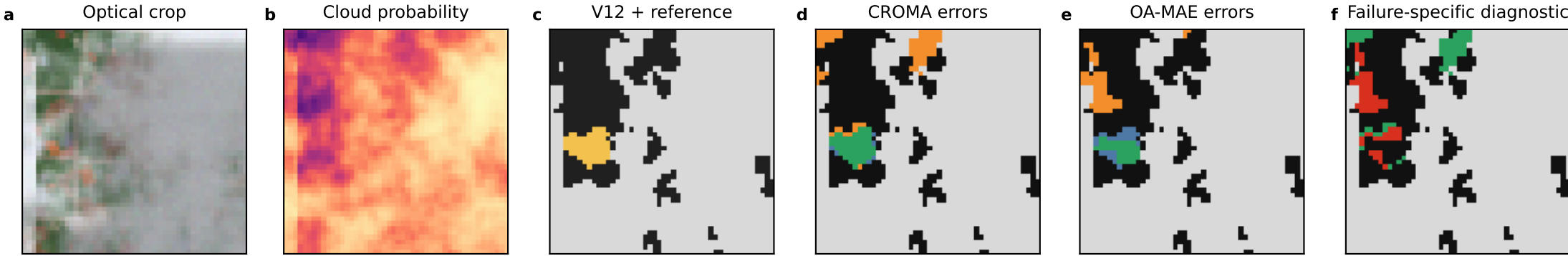


Proposal completeness and independent annotation diagnostics

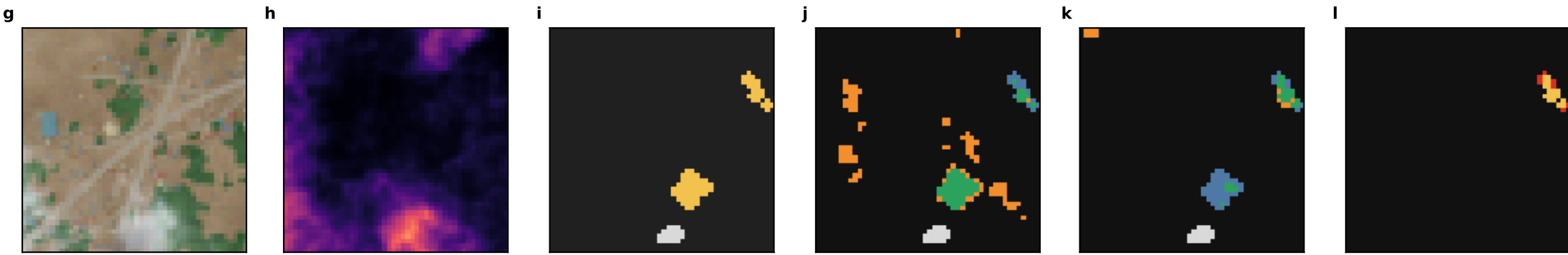


Qualitative taxonomy of failure modes and unresolved cases

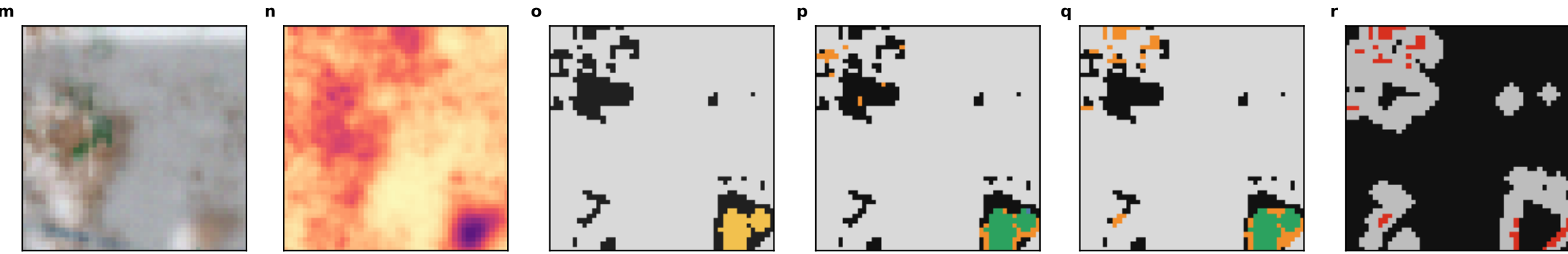
Model regression against CROMA
Kigali | SCN_kig_0066
regular
IoU CROMA 0.333 | OA 0.187 | dIoU -0.147
red: OA wrong / CROMA correct



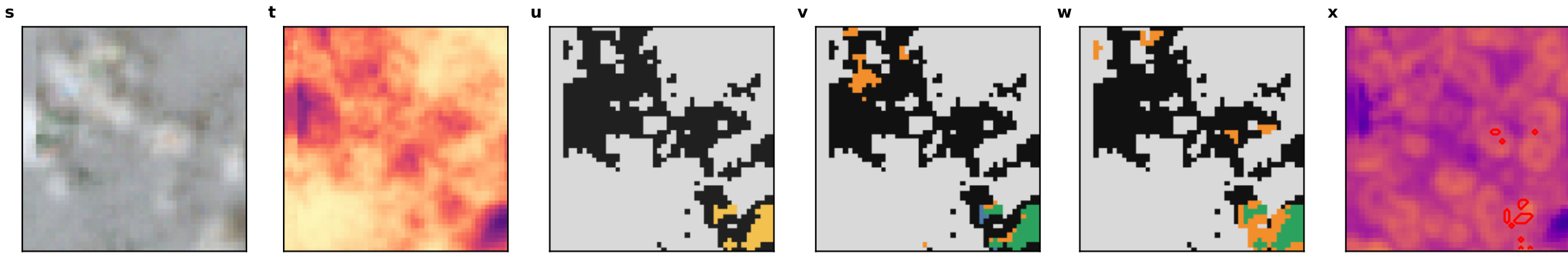
Small-object false negatives
Dar es Salaam | SCN_dar_0064
informal_small_roofs
IoU CROMA 0.202 | OA 0.059 | dIoU -0.143
yellow: small reference; red: missed by OA



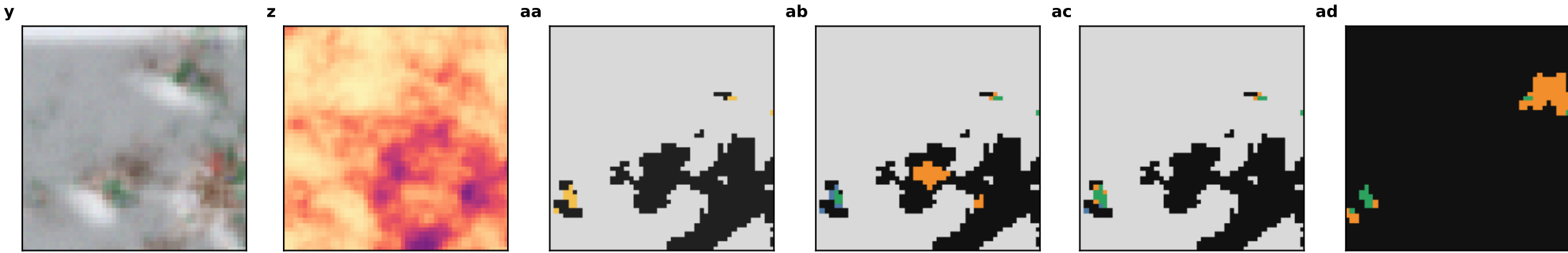
False positives near V12 boundary
Garoua | SCN_gar_0066
regular
IoU CROMA 0.319 | OA 0.325 | dIoU +0.006
gray: V12 boundary; red: OA false positive



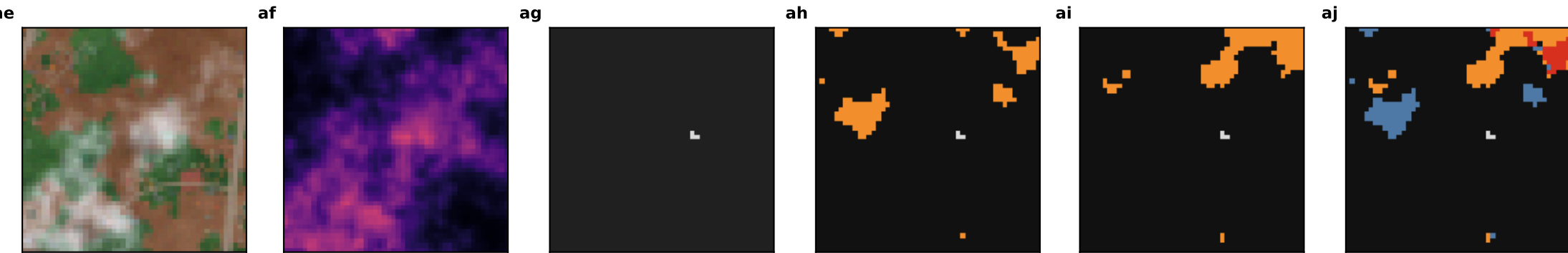
High-gate false positives
Dakar | SCN_dak_0059
texture_complex
IoU CROMA 0.356 | OA 0.353 | dIoU -0.003
effective gate with false-positive contour



Unresolved positive change
Kigali | SCN_kig_0069
regular
IoU CROMA 0.160 | OA 0.261 | dIoU +0.101
green: supported positive; orange: unresolved positive



Shared low-IoU failure
Kigali | SCN_kig_0067
informal_small_roofs
IoU CROMA 0.018 | OA 0.071 | dIoU +0.053
red: both wrong; orange: OA only; blue: CROMA only

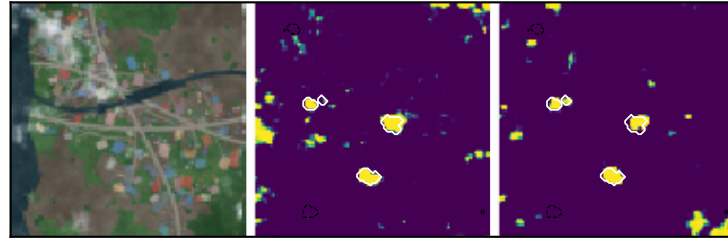


TP FP FN outside V12

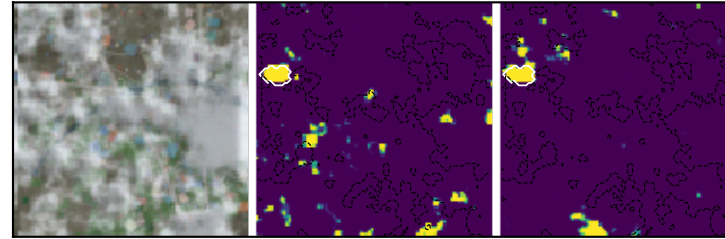
Inter-city qualitative contact sheet - Optical T2 | CROMA | OA-MAE

Douala

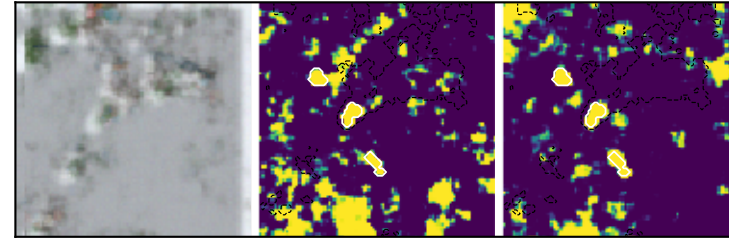
a Low cloud
SCN_dou_0062 | cloud 0.11 | dIoU +0.032



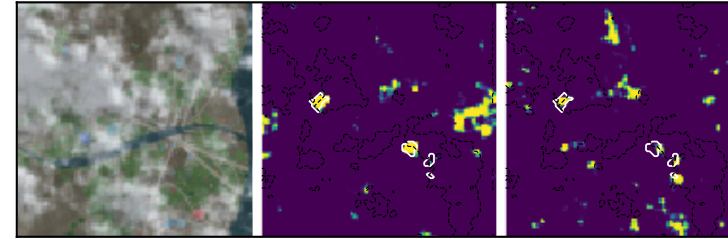
b Median cloud
SCN_dou_0059 | cloud 0.58 | dIoU +0.050



c Severe cloud
SCN_dou_0068 | cloud 0.86 | dIoU +0.059

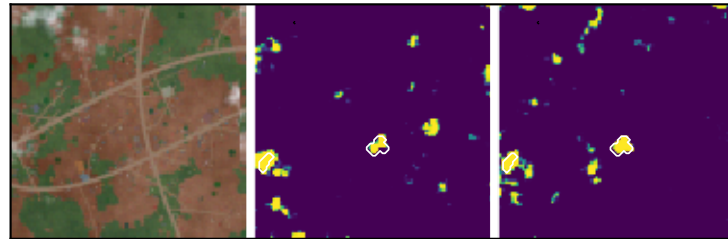


d Worst dIoU
SCN_dou_0067 | cloud 0.48 | dIoU -0.061

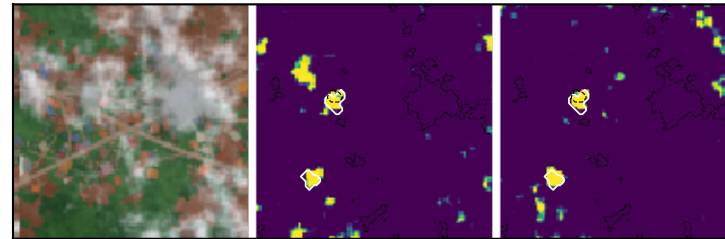


Yaounde

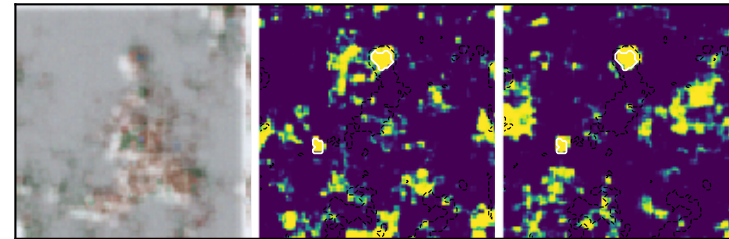
e Low cloud
SCN_yao_0064 | cloud 0.06 | dIoU +0.025



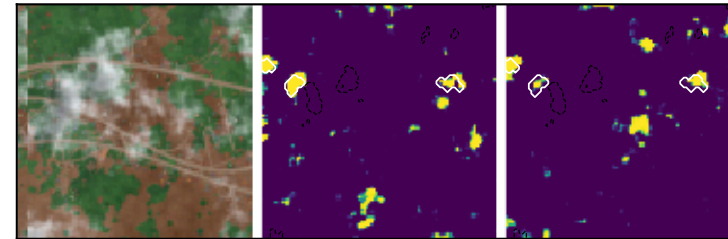
f Median cloud
SCN_yao_0059 | cloud 0.36 | dIoU +0.022



g Severe cloud
SCN_yao_0063 | cloud 0.84 | dIoU +0.005

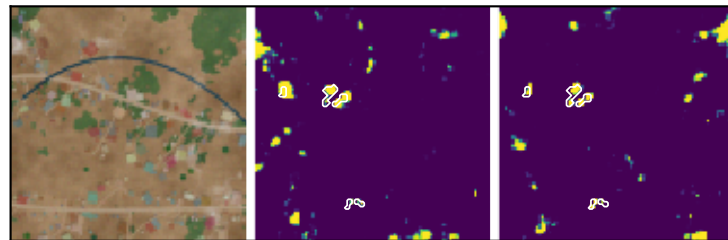


h Worst dIoU
SCN_yao_0067 | cloud 0.24 | dIoU -0.074

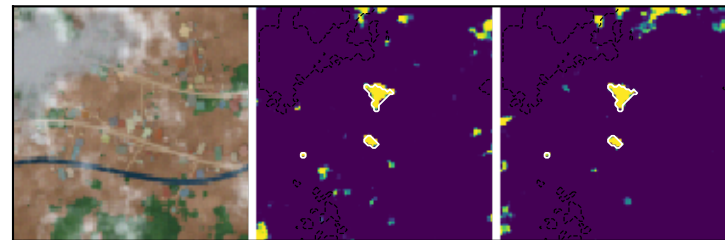


Garoua

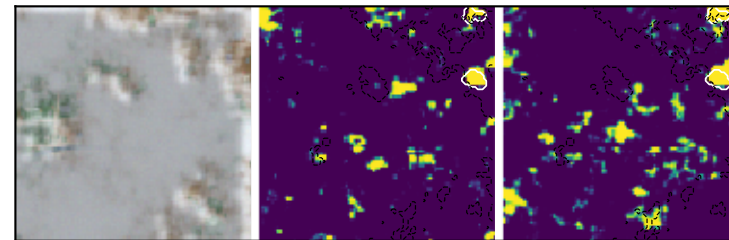
i Low cloud
SCN_gar_0059 | cloud 0.06 | dIoU -0.004



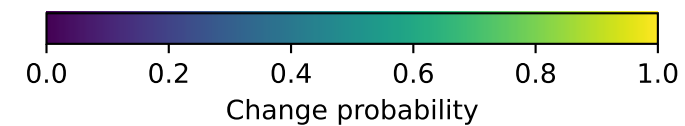
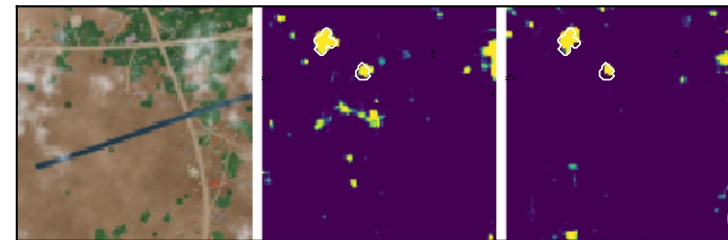
j Median cloud
SCN_gar_0062 | cloud 0.34 | dIoU +0.005



k Severe cloud
SCN_gar_0060 | cloud 0.85 | dIoU +0.005

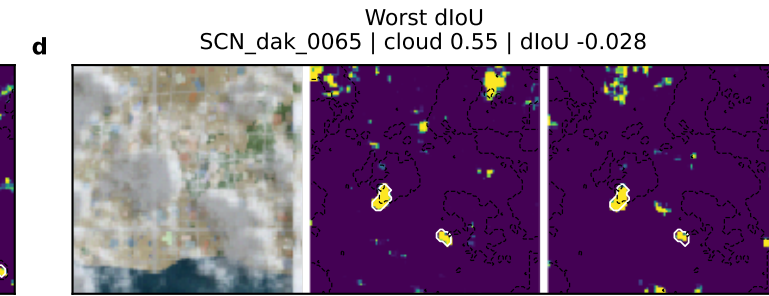
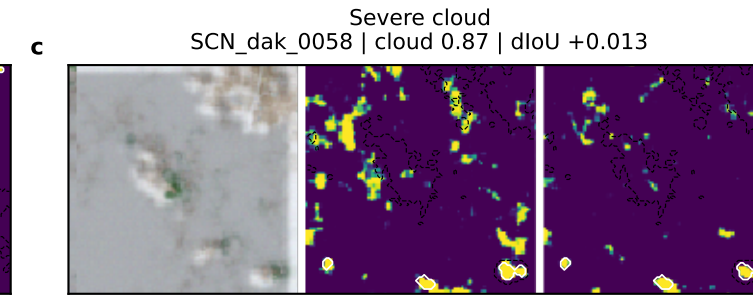
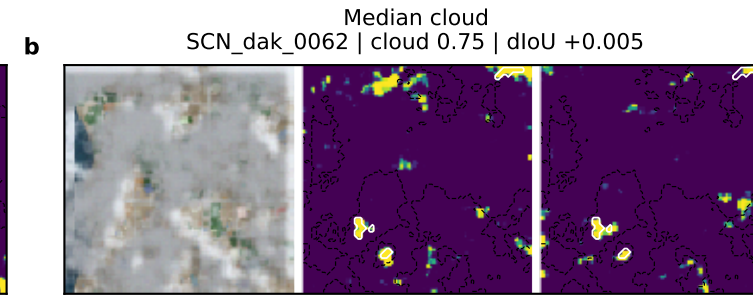
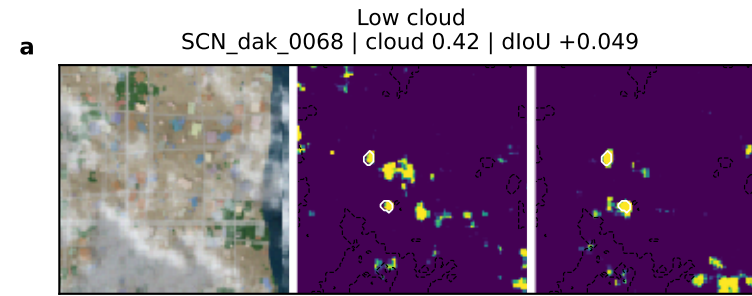


l Worst dIoU
SCN_gar_0058 | cloud 0.15 | dIoU -0.055

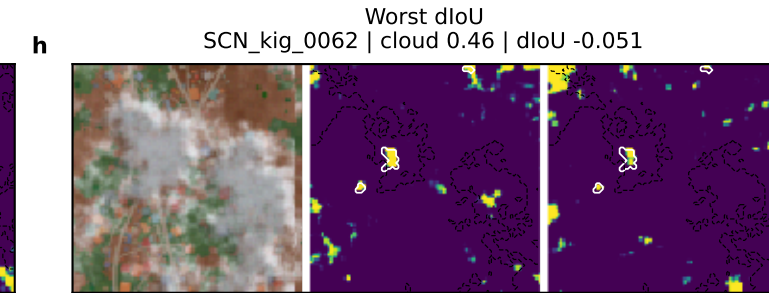
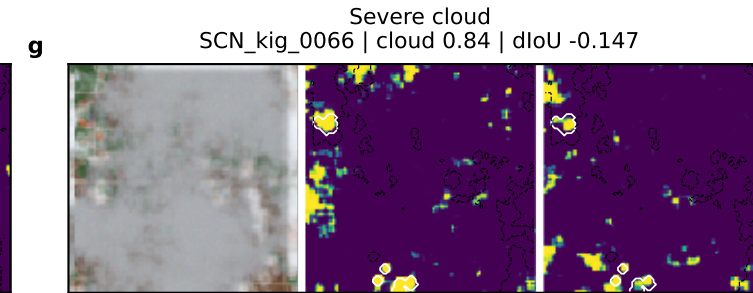
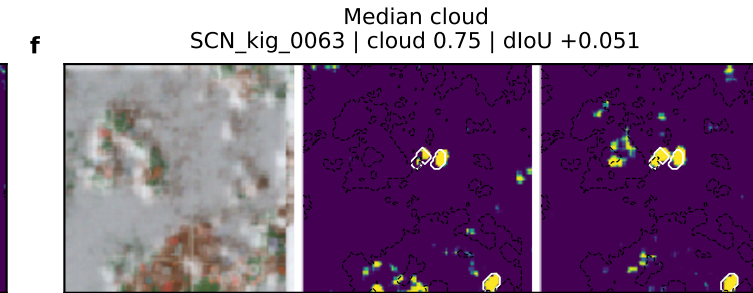
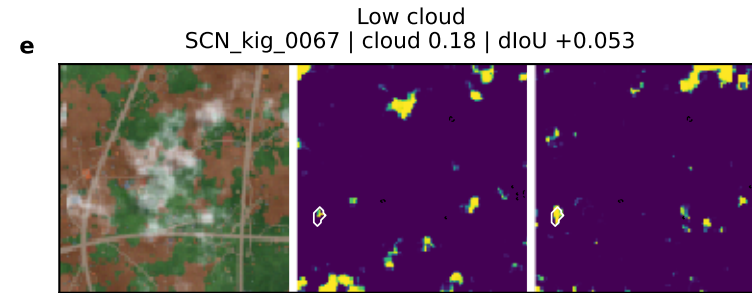


Inter-city qualitative contact sheet - Optical T2 | CROMA | OA-MAE

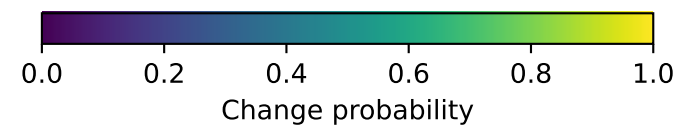
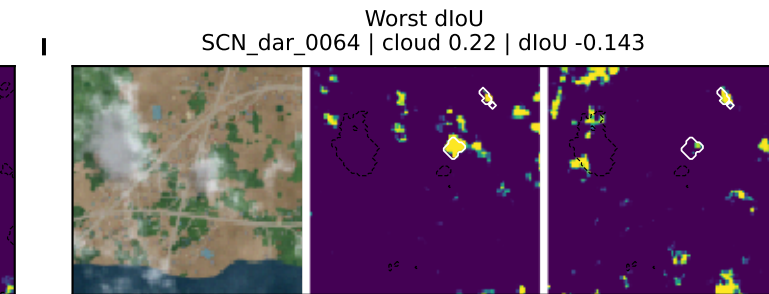
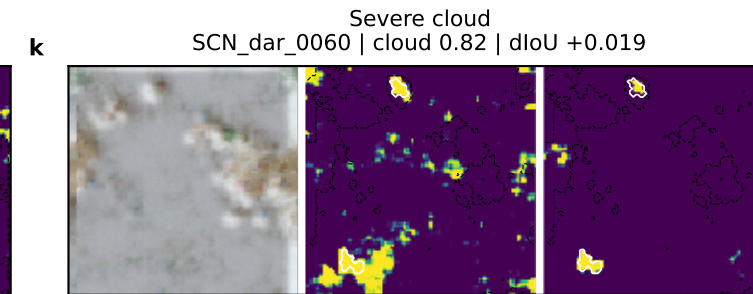
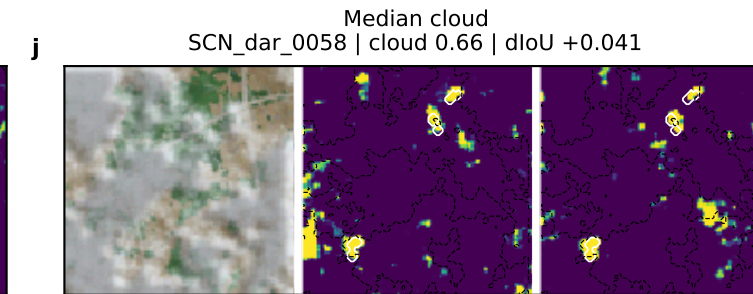
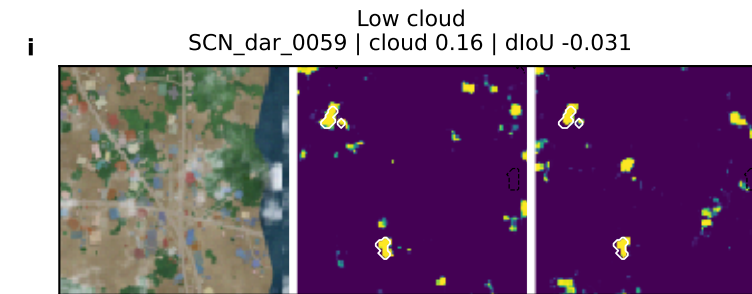
Dakar



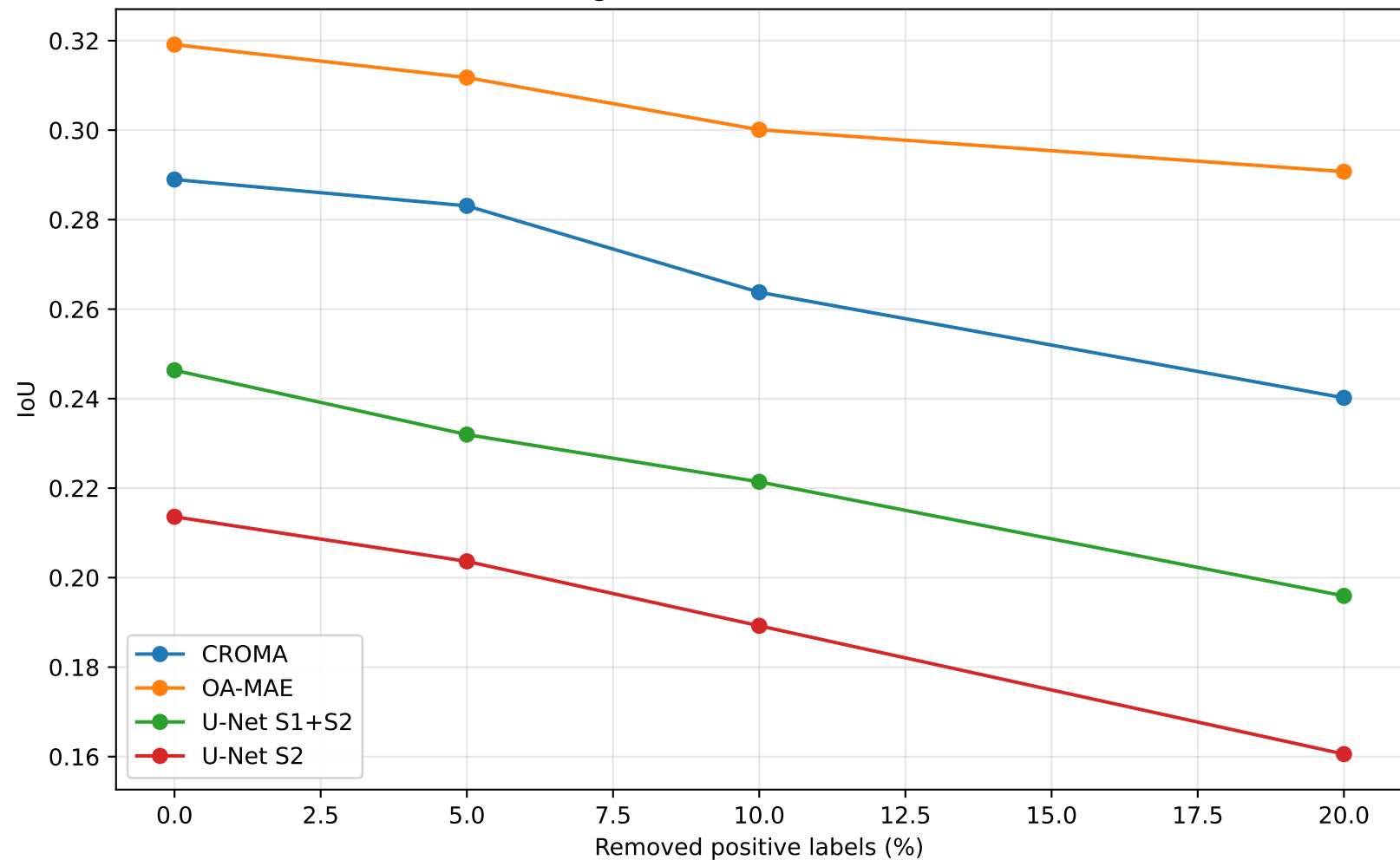
Kigali



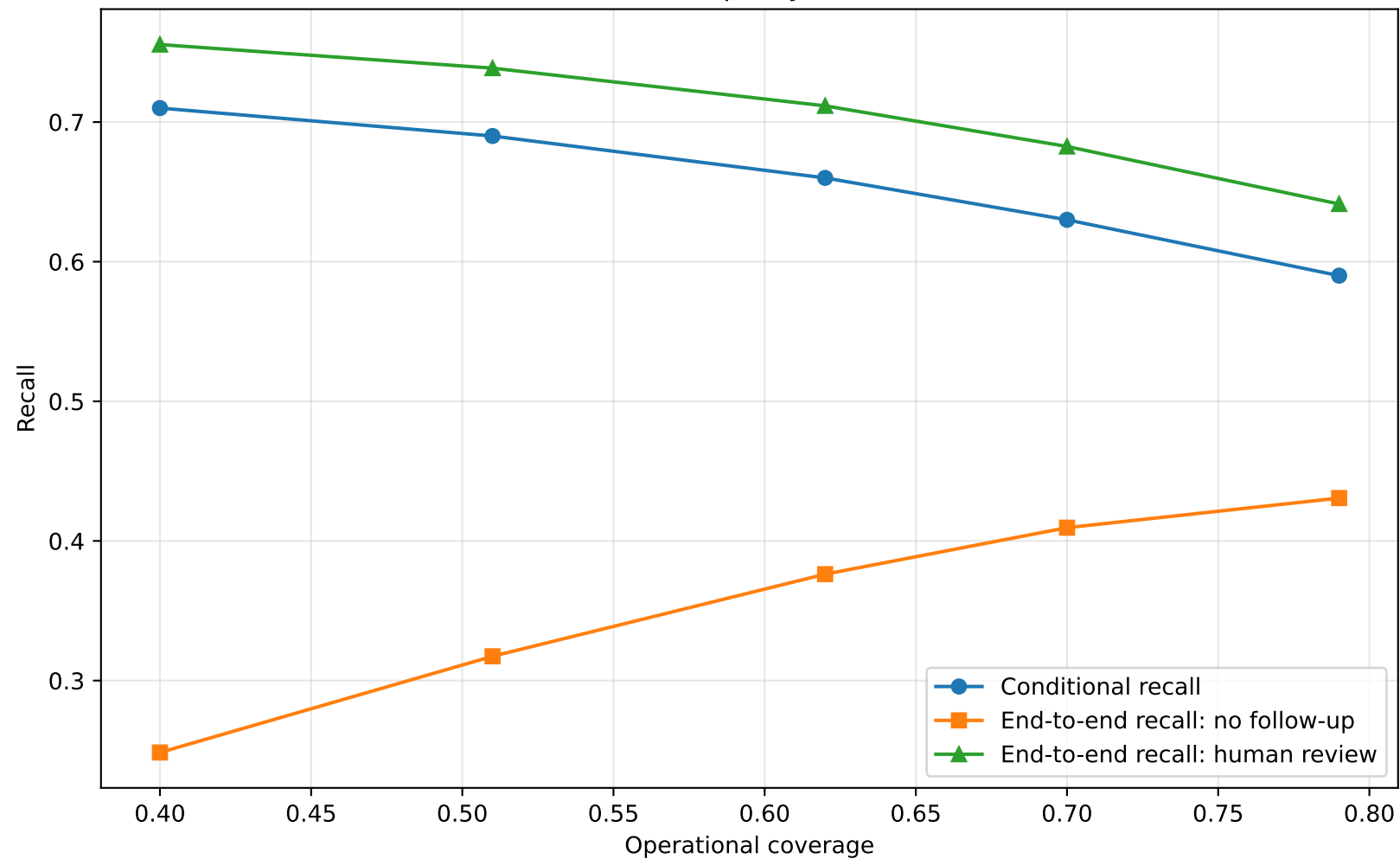
DarEsSalaam



Stage II label-noise robustness



Abstention policy trade-off



OA-MAE gain by cloud-burden bin

